Mathematics — Question Paper

Time: 60 minutes Maximum marks: 40

Attempt all questions. Show your working for Sections B and C.

Section A — Multiple choice (1 mark each)

1. The sum of the roots of $x^2 - 5x + 6 = 0$ is: [1]

(a) 5 (b) 6 (c) −5 (d) 1

2. The degree of $3x^3 - x + 5$ is: [1]

(a) 1 (b) 2 (c) 3 (d) 5

3. The discriminant of $2x^2 - 4x + 3 = 0$ is: [1]

(a) −8 (b) 8 (c) 16 (d) −4

4. The roots of $x^2 + 4x + 4 = 0$ are: [1]

(a) real and equal (b) real and distinct (c) not real (d) irrational

5. If α, β are roots of $x^2 - 7x + 10 = 0$, then $\alpha^2 + \beta^2 =$ [1]

(a) 29 (b) 39 (c) 49 (d) 19

6. A hexagon has how many sides? [1]

(a) 5 (b) 6 (c) 7 (d) 8

7. Area of a rectangle 8 cm × 5 cm is: [1]

(a) 13 cm^2 (b) 26 cm^2 (c) 40 cm^2 (d) 80 cm^2

8. Circumference of a circle of radius 14 cm ($\pi = 22/7$): [1]

(a) 44 cm (b) 88 cm (c) 154 cm (d) 616 cm

9. Diagonal of a square of side 6 cm is: [1]

(a) 6 cm (b) $6\sqrt{2}$ cm (c) 12 cm (d) 36 cm

10. $\triangle ABC \sim \triangle DEF$ with sides in ratio 2 : 3. Ratio of their areas: [1]

(a) 2 : 3 (b) 4 : 9 (c) 3 : 2 (d) 8 : 27

11. A chord of length 8 cm is 3 cm from the centre. The radius is: [1]

(a) 4 cm (b) 5 cm (c) 6 cm (d) 7 cm

12. $\sin^2\theta + \cos^2\theta =$ [1]

(a) 0 (b) 1 (c) 2 (d) $\tan^2\theta$

13. $\sin 30° =$ [1]

(a) 1/2 (b) $\sqrt{3}/2$ (c) 1 (d) 0

14. $1 + \tan^2\theta =$ [1]

(a) $\sec^2\theta$ (b) $\text{cosec}^2\theta$ (c) $\cot^2\theta$ (d) $\sin^2\theta$

15. $\sin 60° \cos 30° + \cos 60° \sin 30° =$ [1]

(a) 0 (b) 1/2 (c) 1 (d) $\sqrt{3}$

Section B — Short answer (3 marks each)

16. Solve: $3x - 7 = 11$. [3]

17. Solve $x^2 - 7x + 12 = 0$. [3]

18. Find the area of a triangle with base 10 cm and height 6 cm. [3]

19. A 13 m ladder reaches 12 m up a wall. How far is its foot from the wall? [3]

20. Evaluate $2\tan^2 45° + \cos^2 30° - \sin^2 60°$. [3]

Section C — Long answer (5 marks each)

21. Solve $2x^2 - 5x - 3 = 0$ by factorisation. [5]

22. Prove $\sin\theta/(1 - \cos\theta) + (1 - \cos\theta)/\sin\theta = 2\,\text{cosec}\,\theta$. [5]